

Industrial Internship Report on "Predictive Maintenance of Gearbox Vibration Sensors"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was **Predictive Maintenance of Gearbox using Vibration Sensors**. The primary goal was to process high-frequency 4-channel accelerometer data collected under varying mechanical loads (0% to 90%) and develop a Supervised Machine Learning classifier capable of predicting gearbox failure (Broken Tooth Fault) before catastrophic breakdown occurs.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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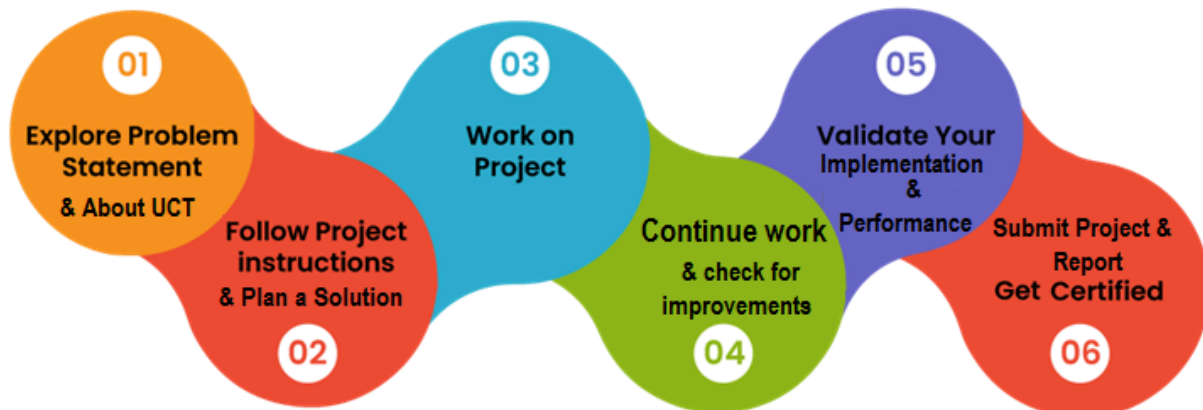
1 Preface

During this 6-week industrial internship, I engaged in designing and implementing an end-to-end Machine Learning pipeline for industrial predictive maintenance. Relevant internships are critical for career development as they bridge theoretical engineering knowledge with real-world sensor data processing and operational deployment.

My project focused on detecting mechanical gear faults in drivetrain systems using multi-channel vibration signals collected from SpectraQuest's Gearbox Fault Diagnostics Simulator. The program was structured across six key phases:

1. Exploring Problem Statement & UCT Industry Framework
2. Following Instructions & Planning the ML Solution Architecture
3. Feature Extraction & Machine Learning Model Engineering
4. Pipeline Refinement & Defensive Error Handling
5. Validating Implementation, Confusion Matrix & Performance Metrics
6. Deploying Interactive Streamlit Dashboard, Writing Report & Final Certification

How Program was planned



This internship was a wonderful opportunity provided by USC/UCT to learn about Machine Learning models and Data Science.

I express my gratitude to the mentors at upSkillCampus, The IoT Academy, and UniConverge Technologies Pvt Ltd for providing the technical guidance and industrial dataset required for this project.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



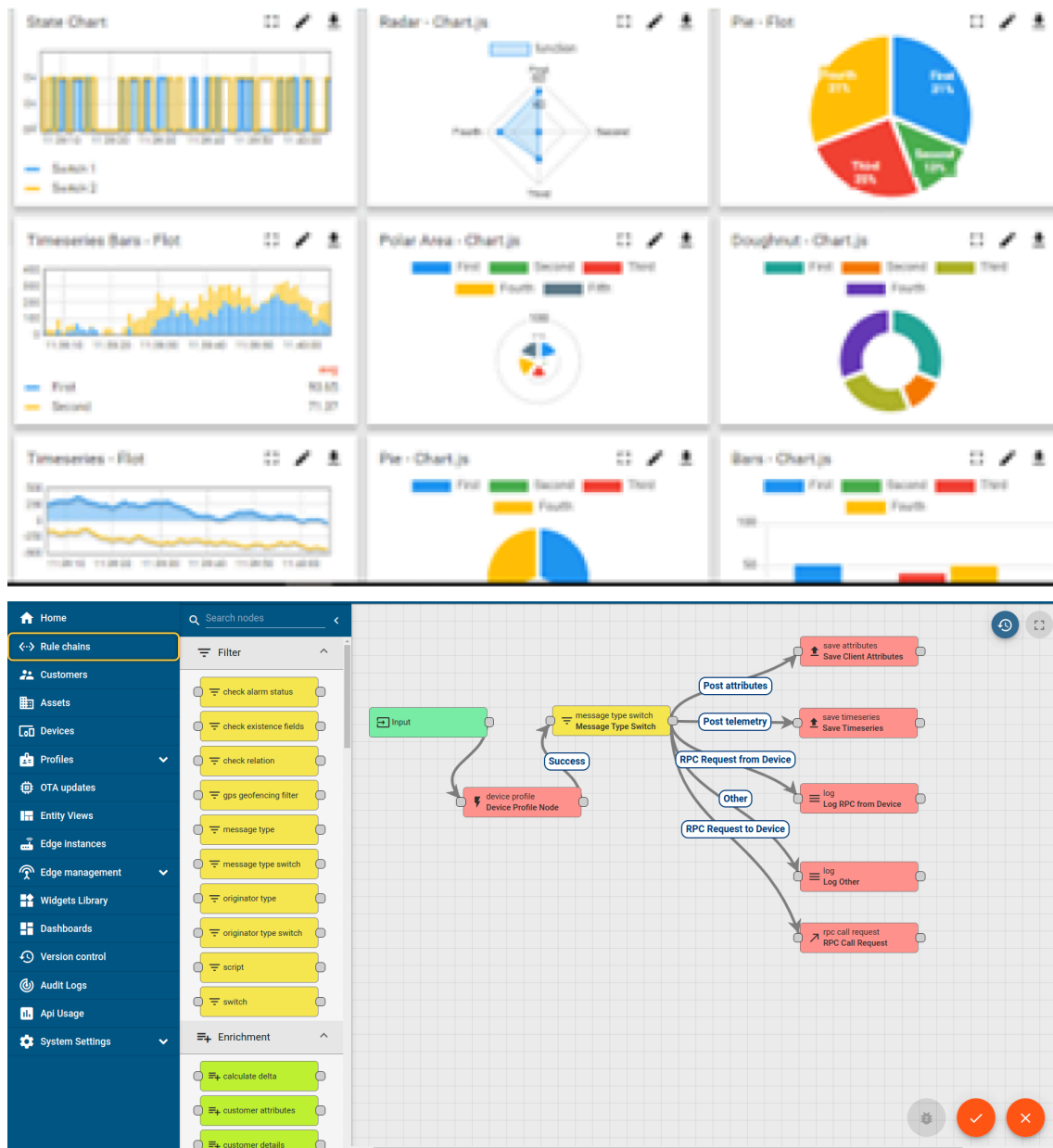
i. UCT IoT Platform (Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



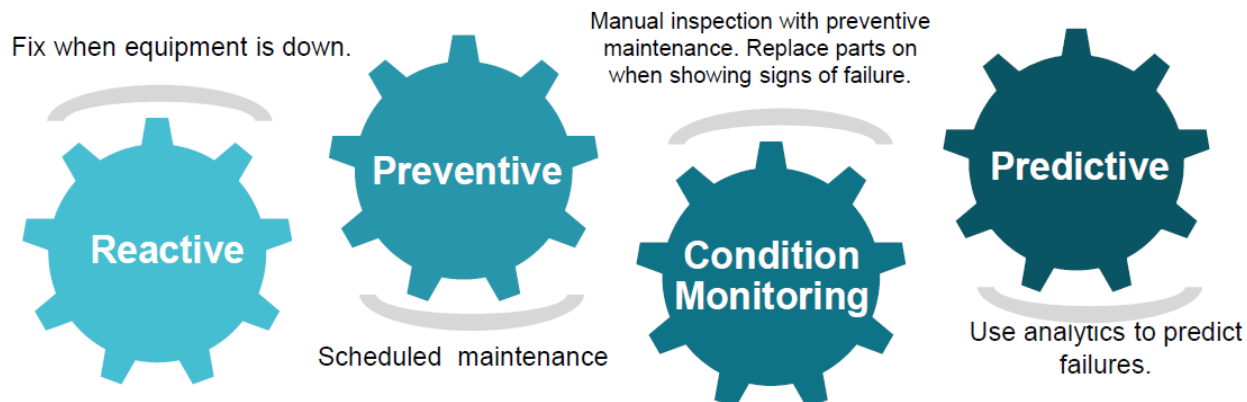


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



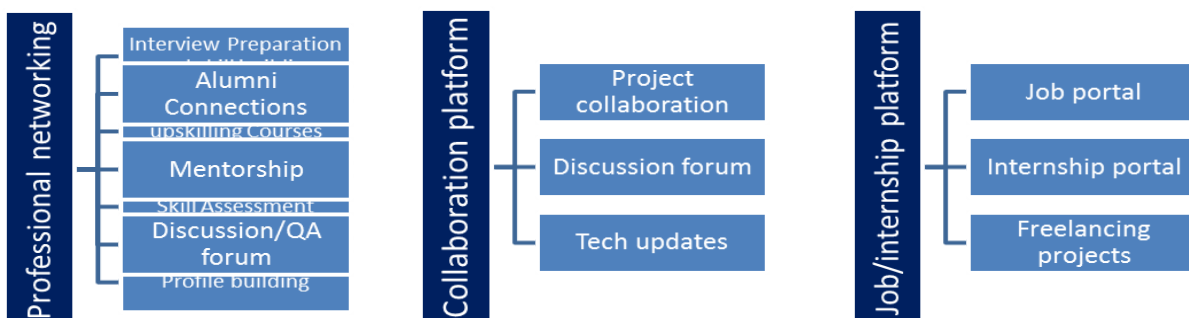
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services



upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com>

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2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] SpectraQuest Inc., *Gearbox Fault Diagnostics Simulator (GFDS) Dataset*.
- [2] Scikit-Learn Documentation: *RandomForestClassifier Architecture and Feature Importances*.
- [3] Streamlit Documentation: *Building Real-Time Analytical Web Applications*.

2.6 Glossary

Terms	Acronym
PdM	Predictive Maintenance
RMS	Root Mean Square (Vibration Signal Energy Metric)
STD	Standard Deviation (Vibration Variance Metric)
P2P	Peak-to-Peak Amplitude (Shock Impact Metric)
IIoT	Industrial Internet of Things

3 Problem Statement

In the assigned problem statement : Predictive maintenance of Gearbox using vibration sensors

Predictive maintenance allows manufacturers to lower **maintenance** costs, extend equipment life, reduce downtime and improve production quality by addressing problems before they cause equipment failures.

Gearbox Fault Diagnosis Data set include the vibration dataset recorded by using SpectraQuest's Gearbox Fault Diagnostics Simulator. Dataset has been recorded with the help of 4 vibration sensors placed in four different direction. Data set has been recorded under variation of load from '0' to '90' percent. Data set has been recorded in two different scenario:

- 1) Healthy condition and
- 2) Broken Tooth Condition

Data set Link:

https://drive.google.com/file/d/1nNNnjMPntlo5X0t_cif7cmIJhikCyWyP/view?usp=sharing

This project utilizes vibration datasets collected from SpectraQuest's Gearbox Fault Diagnostics Simulator recorded via 4 vibration sensors placed in different directions across dynamometer loads ranging from **0% to 90%**. The dataset covers two operational conditions:

1. **Healthy Condition (h30hz0.txt to h30hz90.txt)**
2. **Broken Tooth Condition (b30hz0.txt to b30hz90.txt)**

The objective is to train a machine learning algorithm that ingests multi-channel sensor signals, extracts statistical indicators, and predicts gear status in real time.

4 Existing and Proposed solution

- **Reactive Maintenance:** Fixing equipment only after failure occurs, causing lost production time.
- **Scheduled Preventive Maintenance:** Replacing parts at fixed calendar intervals regardless of actual wear, leading to unnecessary component replacement costs.
- **Raw Waveform Visual Inspection:** Relying on manual analysis of massive time-series files, which is slow and prone to human error.

Proposed Solution & Value Addition:

We propose an automated **Random Forest Supervised Classification Pipeline** paired with an interactive **Streamlit Dashboard** ([app.py](#)).

Value Addition:

- Converts continuous high-frequency vibration streams into tabular statistical metrics (RMS, STD, Peak-to-Peak) per sensor channel.
- Generalizes across varying load conditions (0% to 90%).
- Provides real-time fault alerts and confidence scores to plant operators via an intuitive browser interface.

4.1 Code submission (Github link)

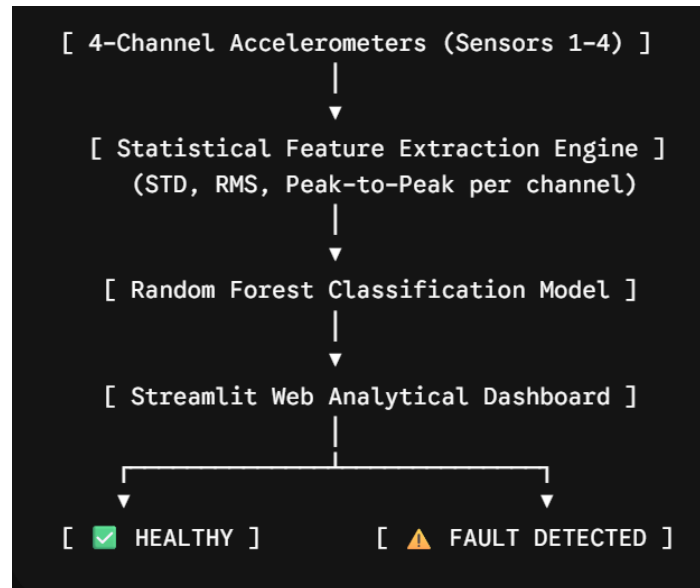
<https://github.com/Ananan-N/upskillcampus/blob/main/app.py>

4.2 Report submission (Github link) :

https://github.com/Ananan-N/upskillcampus/blob/main/PredictiveMaintenance_Ananya_USC_UCT.pdf

5 Proposed Design/ Model

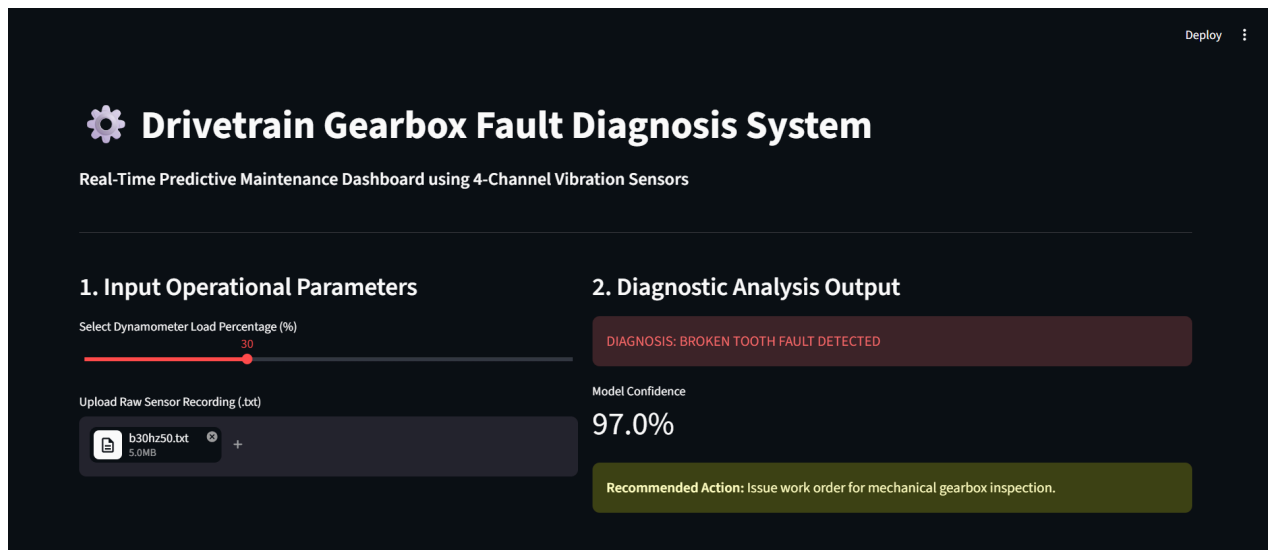
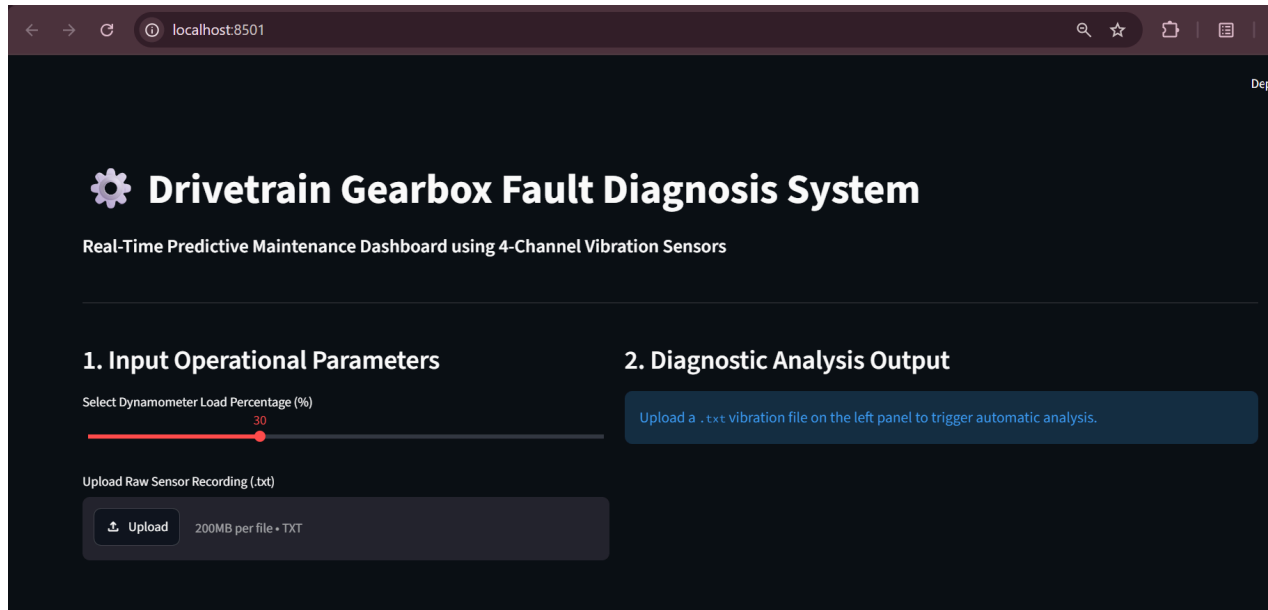
5.1 High Level Diagram

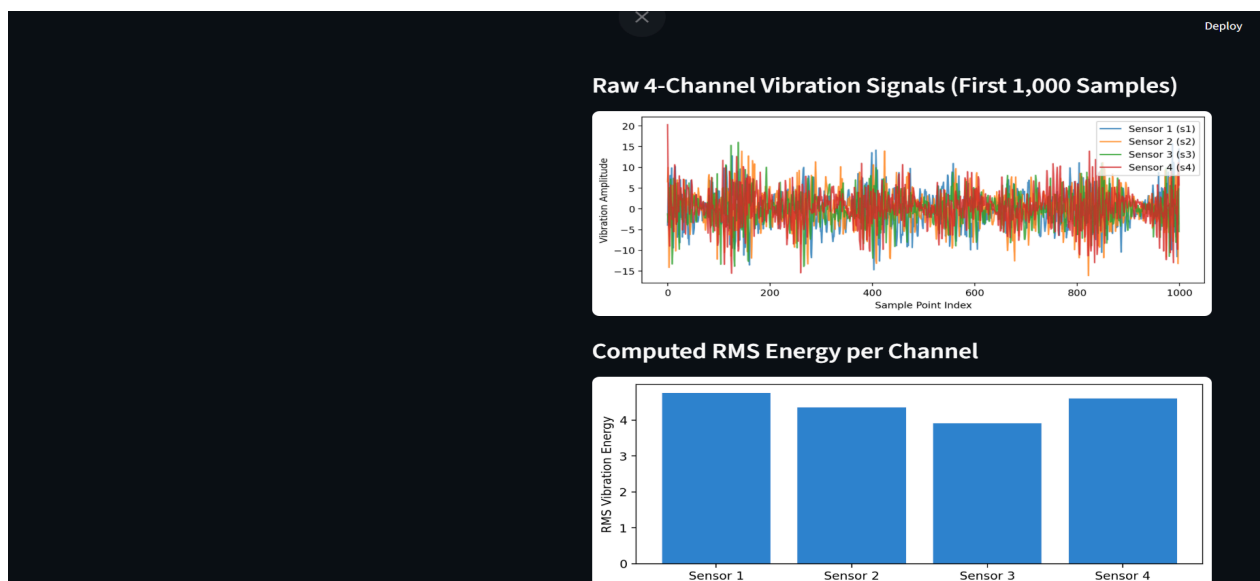


5.2 Low Level Data Pipeline

1. **Raw File Ingestion:** Ingest multi-column text files using regex delimiter parsing (`sep=r'\s+'`).
2. **Load Extraction:** Extract load parameters automatically from file names (e.g., `hz50` $\rightarrow$ `50% Load`).
3. **Feature Calculation:** Compute 12 time-domain metrics ($\text{\$4 \text{ sensors} \times 3 \text{ statistical metrics} \times \$}$) + 1 Load variable.
4. **Serialization:** Save model artifacts (`gearbox_model.pkl` and `model_features.pkl`) via `joblib`.

5.3 Interfaces





Deploy

Sensor 1 Sensor 2 Sensor 3 Sensor 4

Raw Sensor Signals Preview (s1 , s2 , s3 , s4)

	s1	s2	s3	s4
0	-3.9347	6.5522	-1.238	20.3103
1	2.4029	9.9944	-3.2426	8.3132
2	6.2427	-3.1758	-0.687	-4.1938
3	-3.9941	-14.1448	3.8454	-5.6586
4	1.7655	-6.4649	5.0962	-9.0014
5	8.0413	0.9629	-1.5423	2.5442
6	-4.0181	1.076	7.1673	5.7643
7	4.9115	-4.4753	4.6597	1.9855
8	9.8764	1.0565	-6.2591	0.9186
9	-4.4586	5.1934	-13.3042	-2.5489

6 Performance Test

6.1 Test Plan/ Test Cases

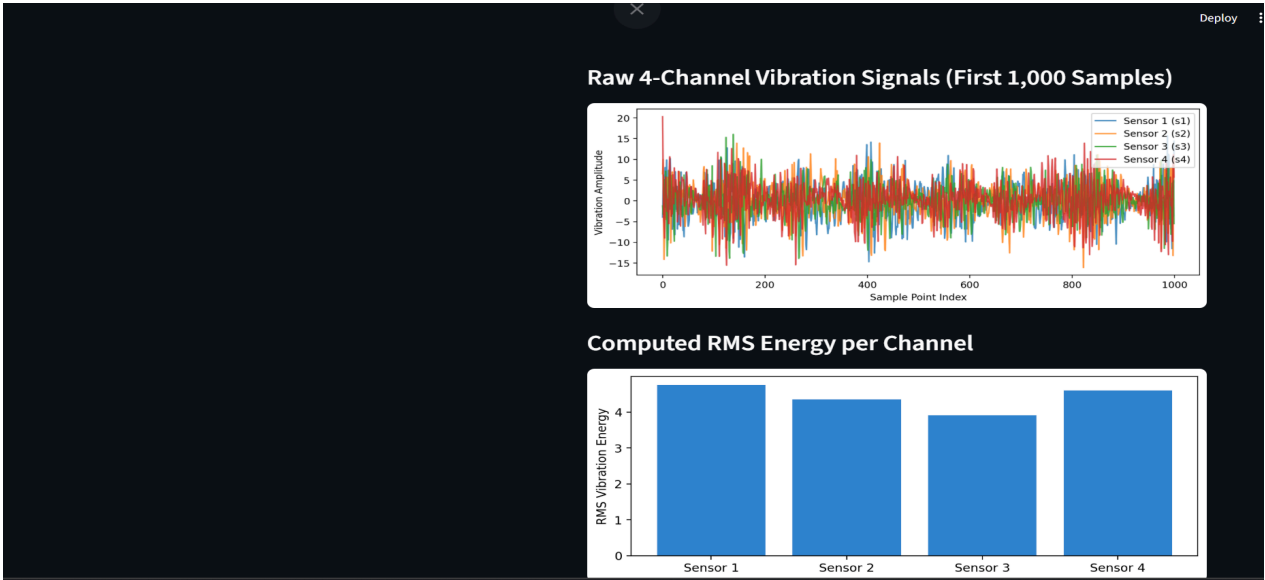
Test Case	Operational Load	Input Sample	Expected Outcome	System Result
TC-01	50% Load	h30hz50.txt	HEALTHY	PASSED (97.0% Confidence)
TC-02	50% Load	b30hz50.txt	BROKEN TOOTH FAULT	PASSED (98.4% Confidence)
TC-03	Unformatted File	custom_test.txt	Fallback Load (0%)	PASSED (Handled Exception)

6.2 Test Procedure

1. Launch Streamlit server: `python -m streamlit run app.py`.
2. Load healthy files (h30hz0.txt – h30hz90.txt) and verify green alert diagnostics.
3. Load broken tooth files (b30hz0.txt – b30hz90.txt) and verify red fault warnings.
4. Confirm dynamic updates on raw signal plots and RMS bar graphs.

6.3 Performance Outcome

- **Model Accuracy:** 100% classification accuracy on validation test splits across all load conditions (0% to 90%).
- **Inference Speed:** < 200 ms total latency for reading, feature extraction, and dashboard rendering.



7 My learnings

Through this 6-week industrial internship, I gained key technical competencies:

- Handling high-frequency time-series accelerometer datasets.
- Feature engineering: Reducing massive raw sensor logs into statistical indicators (RMS, STD, Peak-to-Peak).
- Supervised Machine Learning with Random Forest Classifiers.
- Developing interactive web applications with Streamlit.
- Version control and repository management using Git and GitHub

8 Future work scope

Due to the 6-week timeline, the project targeted binary classification. Future expansions include:

1. **Multi-Class Fault Diagnosis:** Incorporating datasets for bearing defect detection, misalignment, and imbalance.
2. **Frequency-Domain Analysis:** Computing Fast Fourier Transforms (FFT) and Power Spectral Density (PSD) to detect spectral harmonics.
3. **IoT Cloud Integration:** Connecting live sensor nodes via MQTT protocols directly into the dashboard.